Employee Productivity Study

Measuring output, quality and attendance across departments, and testing what is linked to performance

Scope	120 employees, 5 departments, January - June 2026
Records	720 clean employee-month rows (728 raw)
Tools	Python (pandas, scipy, matplotlib), Excel (pivot tables, CORREL)
Data	Synthetic, anonymised IDs, generated for learning

Headline results

Company task completion rate	81.5%
Attendance rate	96.0%
Average quality score	80.8 / 100
Strongest / weakest department (completion)	Operations (82.8%) / Sales (76.8%)
Employees averaging over 20 overtime hours/month	22 of 120
Quality gap, high vs low overtime	-3.3 points

1. Objective and Scope

This study asks four questions: how productive is each department and employee, how has output changed over six months, which factors move with productivity (overtime, attendance, tenure, training), and who may need support or recognition.

Deliverables

- Cleaned employee-month dataset and an employee score table (CSV)
- A fair productivity index that compares people only within their own department
- Trend, department, correlation and training analysis with charts
- Recommendations and the limits of what the data can prove

A caution before reading the results

Productivity numbers describe measurable output only. They miss mentoring, collaboration and work that is hard to count. The results here are meant to guide support and process changes, not to be the sole basis for pay or dismissal decisions. The data is synthetic.

2. Dataset Description

Column	Meaning
Employee_ID / Department / Level	Anonymised ID, team, and Junior / Mid / Senior
Tenure_Months	Months with the company
Month	January to June 2026
Hours_Worked / Overtime_Hours	Total hours in the month, and hours beyond standard
Tasks_Assigned / Tasks_Completed	Workload and finished items
Days_Present / Working_Days	Attendance in the month
Quality_Score	Review or audit score, 0-100
Training_Hours	Training received that month

Sample rows (cleaned)

ID	Dept	Month	Hours	Assigned	Done	Present	Quality
E1112	Operations	Apr	176	224	192	21/22	83
E1082	Finance	May	180	133	118	21/21	78
E1100	Engineering	Feb	161	52	44	19/20	81
E1098	Operations	Jun	184	230	195	20/22	82
E1033	Finance	Feb	143	123	104	17/20	82
E1025	Engineering	Apr	177	61	41	22/22	81
E1007	Sales	Mar	196	122	84	21/22	80
E1103	Engineering	Jun	183	70	54	20/22	80

3. Data Cleaning

Five problems were found and fixed before any metric was calculated.

Issue	Count	Fix
Duplicate rows	8	Removed
Inconsistent department text	30	Trimmed and converted to Title Case
Tasks completed greater than assigned	6	Capped at assigned
Days present greater than working days	5	Capped at working days
Missing quality score	12	Filled with that employee's average quality

Result: 728 raw rows became 720 clean rows (120 employees x 6 months = 720 expected). Capping and filling are assumptions; in practice each flagged record would be checked with the team lead or HR system.

4. Method

Why a within-department index. Departments do different work: a support agent closes many small tickets per hour, an engineer finishes fewer, larger tasks. Comparing raw tasks per hour across departments would just rank departments by task size. So each employee is scored against colleagues in the same department.

Metrics per employee (6-month totals).

- Completion rate = tasks completed / tasks assigned
- Tasks per hour = tasks completed / hours worked
- Attendance rate = days present / working days
- Quality score = average of monthly scores

Productivity index. Each metric is converted to a z-score within the department, then combined with weights of 35% completion, 30% tasks per hour, 25% quality and 10% attendance, and rescaled so 50 is the department average (10 points is roughly one standard deviation of a single metric). The weights are a judgment call; changing them changes rankings, so they should be agreed with management before use.

5. Analysis

5.1 Monthly trend

Month	Completion %	Attendance %	Quality	Avg overtime (h)
Jan	81.0	95.9	80.6	12.7
Feb	81.4	95.4	80.4	11.6
Mar	79.0	96.4	80.5	12.5
Apr	82.0	96.3	81.2	12.9
May	82.4	96.0	80.8	11.9
Jun	83.2	96.0	81.4	12.6

Completion was lowest in Mar (79.0%). April-June averaged 82.5% against 80.5% for January-March.

5.2 Department comparison (raw metrics, comparable across teams)

Department	Employees	Completion %	Attendance %	Quality	Avg overtime (h/mo)
Operations	22	82.8	94.9	80.2	11.3
Support	30	82.5	96.2	81.3	12.5
Finance	17	82.0	95.8	79.9	13.2
Engineering	28	79.0	96.4	81.2	12.8
Sales	23	76.8	96.4	81.1	12.0

5.3 Seniority

Level	Employees	Completion %	Quality	Mean index
Junior	50	78.9	79.3	44.5
Mid	48	81.2	81.7	53.2
Senior	22	83.0	82.5	55.6

Index values across levels are shown for reference; the index is built within department, not within level.

5.4 What moves with productivity (correlations)

Pearson correlation across employees (n = 120). r ranges from -1 to +1; values near 0 mean little linear relationship. Correlation does not show cause.

Pair	r	Significance
Attendance rate vs completion rate	+0.16	p = 0.091
Average overtime vs quality score	-0.31	p < 0.001
Average overtime vs tasks per hour (dept-adjusted)	-0.27	p = 0.003
Tenure vs productivity index	+0.21	p = 0.023
Quality score vs completion rate	+0.77	p < 0.001

5.5 Overtime

Employees grouped by average monthly overtime:

Group	Employees	Completion %	Quality	Attendance %	Mean index
Low (under 10h/mo)	62	81.4	81.6	96.0	52.4
Moderate (10-20h)	36	81.5	81.1	95.8	51.8
High (over 20h)	22	76.8	78.2	96.2	40.3

Quality for the high-overtime group (22 employees) is 78.2 against 81.6 for the low group (62), a gap of 3.3 points (Welch t-test, p = 0.002). The pattern suggests that sustained overtime is associated with lower quality, but employees are not randomly assigned overtime, so this does not prove overtime is the cause.

5.6 Training

56 employees received training during April-June. Completion rate before and after, trained vs not trained:

Group	Jan-Mar %	Apr-Jun %	Change (pts)
Trained	80.3	85.0	+4.7
Not trained	80.5	80.2	-0.3

The trained group improved by +5.0 points more than the untrained group (a difference-in-differences estimate). Caveat: if training was offered to employees who were already motivated or on a rising path, part of this gain is selection, not training.

6. Top and Bottom Performers

Ranked by productivity index (50 = department average). IDs only; a low score is a prompt for a conversation, not a verdict.

Top 10

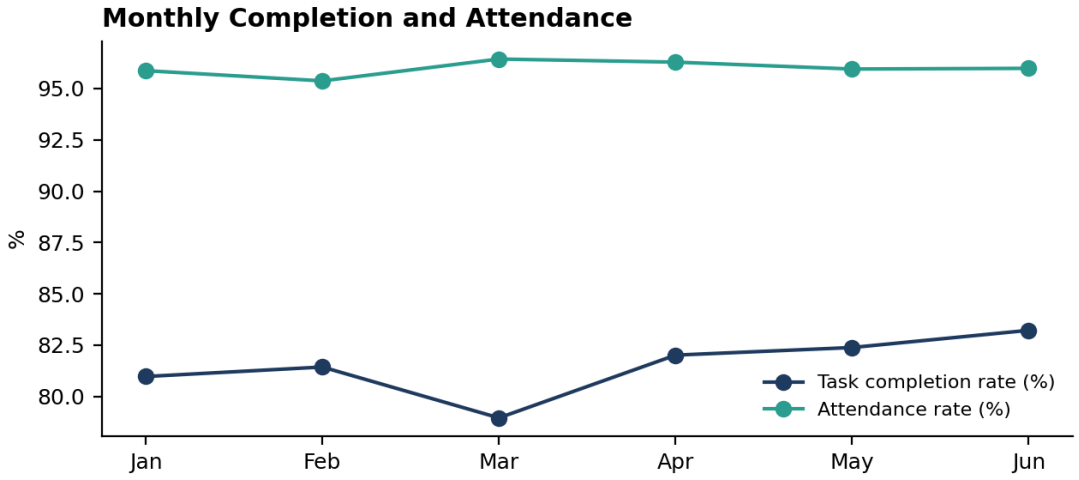
ID	Department	Level	Index	Completion %	Quality	Avg OT (h)
E1014	Support	Senior	89.7	96.3	90.7	8.1
E1043	Support	Mid	82.8	96.0	91.9	17.5
E1083	Sales	Senior	82.7	86.4	88.6	7.0
E1068	Engineering	Mid	82.0	89.2	88.4	4.4
E1040	Engineering	Junior	77.1	88.9	86.5	3.9
E1030	Operations	Mid	76.3	87.7	82.9	5.0
E1004	Finance	Junior	75.1	90.8	86.3	9.1
E1065	Sales	Mid	74.1	86.4	83.5	10.6
E1114	Operations	Mid	73.3	87.7	86.1	7.2
E1017	Support	Senior	71.8	90.1	87.7	9.9

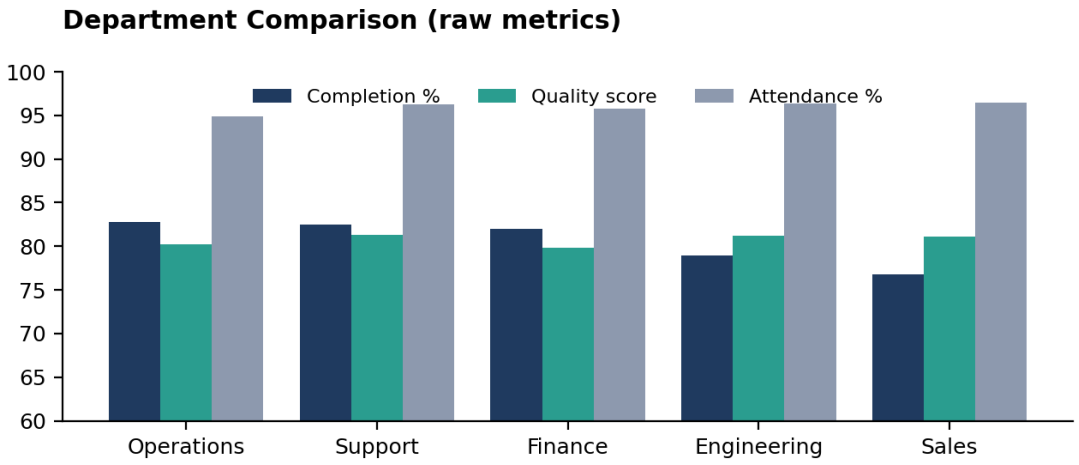
Bottom 10

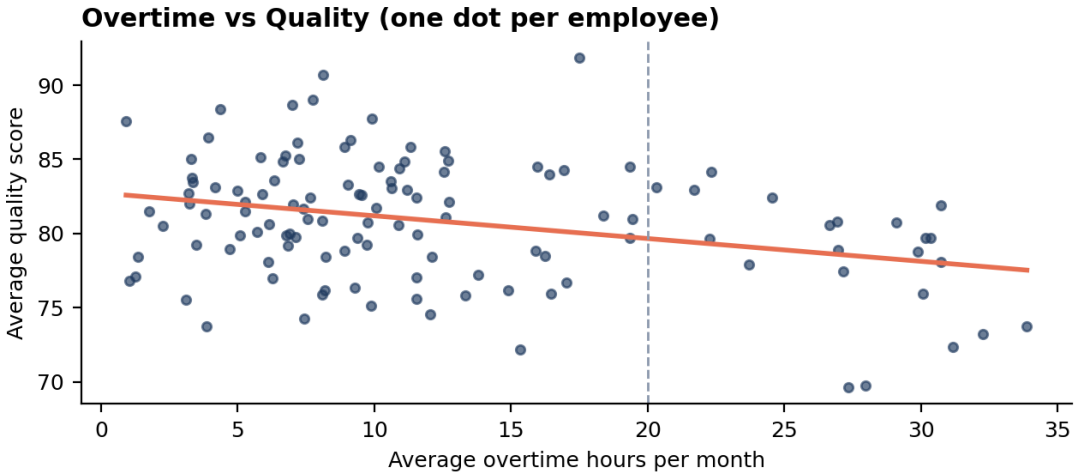
ID	Department	Level	Index	Completion %	Quality	Avg OT (h)
E1057	Support	Junior	25.7	76.5	69.6	27.3
E1120	Support	Senior	25.6	73.3	76.2	14.9
E1044	Finance	Mid	25.6	72.7	79.7	30.4
E1077	Engineering	Junior	24.7	71.5	74.3	7.4
E1113	Engineering	Junior	24.5	71.7	76.4	9.3
E1106	Operations	Mid	15.4	72.7	75.9	8.1
E1066	Finance	Junior	15.1	70.7	69.8	28.0
E1003	Engineering	Junior	14.5	66.9	73.2	32.3
E1093	Support	Junior	13.7	69.3	72.3	31.2
E1102	Sales	Junior	12.9	65.2	72.2	15.3

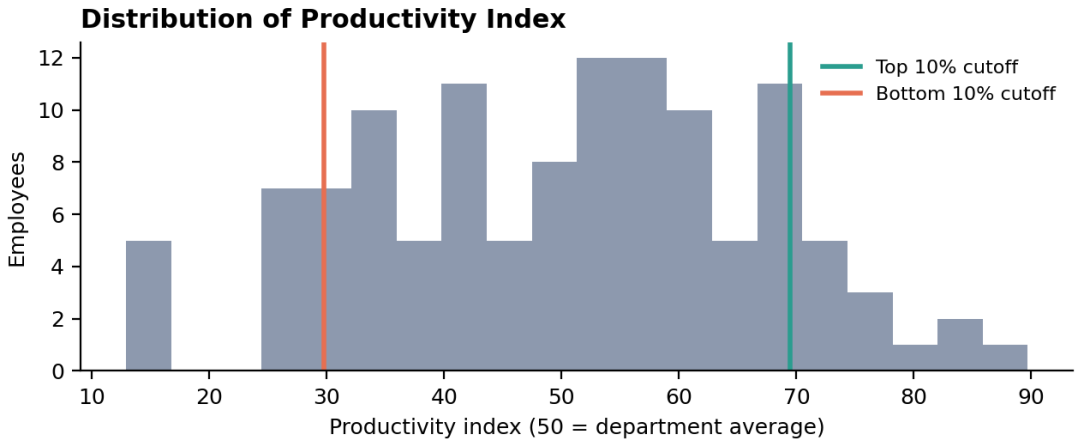
The bottom 10 average 20.4 overtime hours a month against 12.4 for all employees. Departments most represented in the bottom 10: Support (3), Engineering (3), Finance (2). Top 10: Support (3), Sales (2), Engineering (2).

7. Visualizations








Distribution of Productivity Index
Top 10% cutoff
Bottom 10% cutoff
Employees
0
2
4
6
8
10
12
10
20
30
40
50
60
70
80
90
Productivity index (50 = department average)

8. Recommendations

1. Review chronic overtime. 22 employees average over 20 overtime hours a month, and that group scores 3.3 quality points below the low-overtime group. Rebalance workload or add capacity for these people, and consider an overtime alert at 20 hours a month.

2. Expand training, then measure it properly. The trained group improved +5.0 points more than the rest. Roll out to the remaining employees in a staggered way so the next round can be compared fairly.

3. Support, do not punish, the bottom decile. Hold one-to-one reviews with the bottom 10 to check workload, tools and overtime before drawing conclusions. The bottom 10 include Support (3), Engineering (3), so start there.

4. Investigate the Mar dip. Completion fell to 79.0% that month for the whole company, which points to a shared cause (a system change, staffing gap or demand spike) rather than individual performance.

5. Recognise top performers. The top 10 are all above an index of 72; ask what they do differently and share practices.

Metrics to track next quarter: completion rate, share of employees over 20 overtime hours, quality score, and the index gap between trained and untrained staff.

9. Limitations

- Synthetic data; real data has seasonality, role changes, leave and uneven task difficulty.
- Six months is short; trends and the training effect could be noise.
- Weights in the index are a judgment call, and outputs not counted by the system are invisible.
- Correlations and group gaps show association, not cause.
- Within-department scoring means a strong team and a weak team both average 50.

Appendix: Python script (core analysis)

```
import pandas as pd, numpy as np
from scipy import stats
d = pd.read_csv("employee_productivity_raw.csv").drop_duplicates()
d["Department"] = d.Department.str.strip().str.title()
d["Tasks_Completed"] = d[["Tasks_Completed","Tasks_Assigned"]].min(axis=1)
d["Days_Present"]    = d[["Days_Present","Working_Days"]].min(axis=1)
d["Quality_Score"]   = d.Quality_Score.fillna(
    d.groupby("Employee_ID").Quality_Score.transform("mean"))

e = d.groupby("Employee_ID").agg(
    Department=("Department","first"), Hours=("Hours_Worked","sum"),
    OT=("Overtime_Hours","mean"), Assigned=("Tasks_Assigned","sum"),
    Done=("Tasks_Completed","sum"), Present=("Days_Present","sum"),
    WD=("Working_Days","sum"), Quality=("Quality_Score","mean"))
e["Completion"] = e.Done / e.Assigned * 100
e["TPH"]        = e.Done / e.Hours
e["Attend"]     = e.Present / e.WD * 100

z = lambda s: (s - s.mean()) / s.std()
for c in ["Completion","TPH","Quality","Attend"]:
    e["z_"+c] = e.groupby("Department")[c].transform(z)
w = dict(Completion=.35, TPH=.30, Quality=.25, Attend=.10)
raw = sum(w[c] * e["z_"+c] for c in w)
e["Index"] = 50 + 10 * raw / np.sqrt(sum(v**2 for v in w.values()))

r, p = stats.pearsonr(e.OT, e.Quality)
hi, lo = e[e.OT > 20].Quality, e[e.OT <= 10].Quality
t = stats.ttest_ind(hi, lo, equal_var=False)
```

Excel equivalents

- Completion rate: =Completed / Assigned; attendance: =DaysPresent / WorkingDays
- Z-score within department: =(X - AVERAGEIFS(range, DeptRange, Dept)) / STDEV of that department (use a helper column or pivot)
- Correlation: =CORREL(OvertimeRange, QualityRange)
- Department summary: PivotTable, Rows = Department, Values = average completion, quality, attendance